Create a Load Balanced VM Scale Set in Azure SOP

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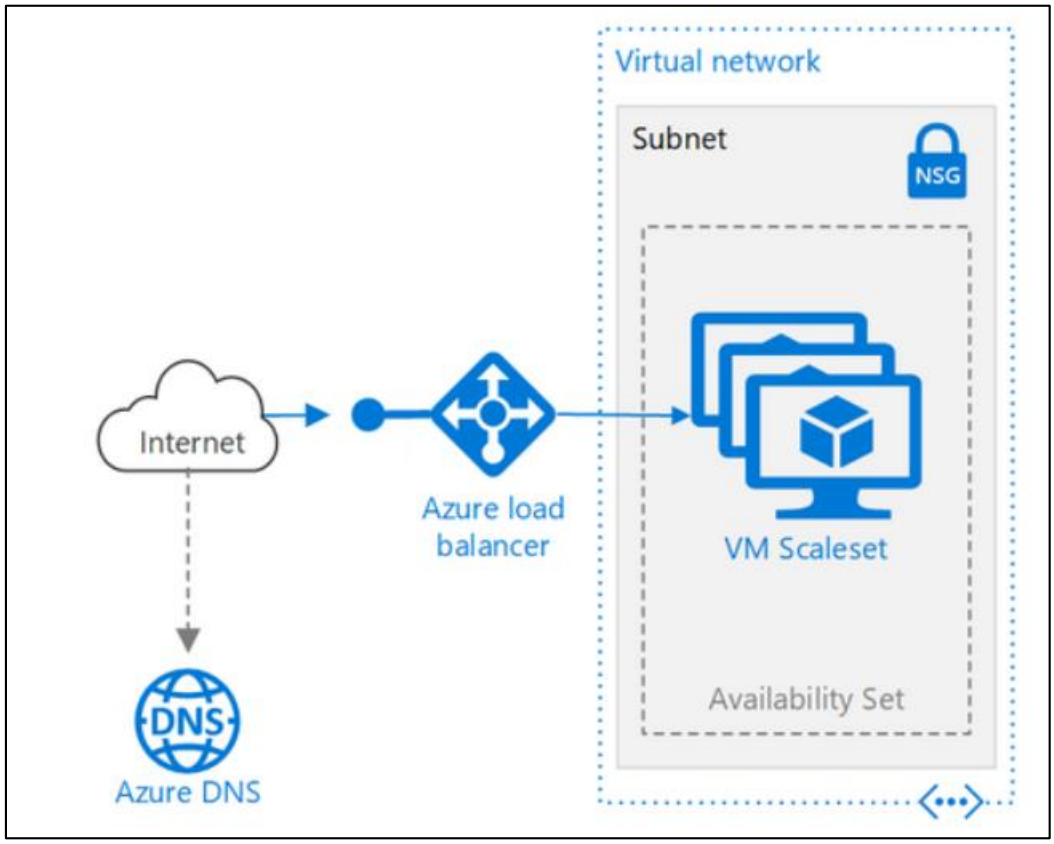
1. Create a Load Balanced VM Scale Set in Azure

1.1 Description

Virtual machine scale sets provide redundancy and improved performance, applications are typically distributed across multiple instances. Customers may access your application through a load balancer that distributes requests to one of the application instances. If you need to perform maintenance or update an application instance, your customers must be distributed to another available application instance. To keep up with additional customer demand, you may need to increase the number of application instances that run your application.
Azure virtual machine scale sets provide the management capabilities for applications that run across many VMs, automatic scaling of resources, and load balancing of traffic.

1.2 Architecture Diagram

The diagram below displays a visual representation of the application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Create a Load Balancer:
    a. Go to azure portal, search for **load balancer** and click **create**.
    b. Provide the following details, please note that the values given here are sample ones:
        i. Resource Group: Your Resource Group name
        ii. Name: labloadbalancer
        iii. Region: **Central US**
        iv. Type: Public
        v. SKU: Standard
        vi. Public IP Address: Create New
        vii. Public IP address name: labpublicip
        viii. Availability Zone: Zone-redundant
    c. Leave other fields with their default values and click **Create**.

2. Create virtual machine scale set:
    a. Type **Scale set** in the search box, select **Virtual machine scale sets** and click **Create**.
        i. Resource Group: Your Resource Group Name
        ii. Scale Set Name: labvmss
        iii. Availability Zone:1
        iv. Image: Windows Server 2019 Datacenter
        v. Size: Standard_B2ms
    b. Provide username and password
    c. Go to Networking Blade.
    d. Virtual Network: Create new Virtual network
        i. Name: labvnet
        ii. Resource Group: your Resource Group
        iii. Replace the name of default subnet with the name labsubnet
        iv. Click **OK**
    e. Network Interface: Create new Nic
        i. Name: labnic
        ii. Subnet will be selected as labsubnet
        iii. NIC network security group: Advanced
        iv. Network Security Group: create new and give it a name lbnsg
        v. Public IP address: Enabled
        vi. Click **Create**
        vii. Now select the default nic (labvnet-nic01) and click **Delete**
    f. Load Balancing: No
    g. Go to **Management Blade** and select **Boot Diagnostics**: **Off**
    h. Leave rest of the fields with their default values and click **Create**.

3. Configure the Network Security Group for port 80:
    a. Go to the lbnsg **Network Security Group**.
    b. Go to **Inbound Security Rules**
    c. Click **Add**
    d. Source: Service Tag
    e. Source Service Tag: Internet
    f. Source Port Ranges: *
    g. Destination: Any
    h. Destination Port Ranges: 80

i. Protocol: TCP
j. Action: Allow
k. Priority: 100
l. Name: Port_80
m. Click **Add.**

4. Configure Load Balancing on VMSS:
   a. Go to the labloadbalancer you created earlier.
   b. Go to the **Backend Pools** and click **Add**:
      i. Name: labbackendpool
      ii. Virtual Network: labvnet
      iii. Associated to: Virtual Machine Scale set
      iv. From VMSS drop down, select labvmss
      v. From IP address, select the default IP of labvmss
      vi. Click **Add**.
   c. Go back to the load balancer and click **Health Probe** > **Add**:
      i. Name: labhealthprobe
      ii. Protocol: HTTP
      iii. Port: 80
      iv. Leave other fields with their default values and click **OK**.
   d. Go to the load Balancing Rules, click **Add**:
      i. Name: labloadbalancerule
      ii. Frontend IP Address: selected by default
      iii. Protocol: TCP
      iv. Port: 80
      v. Backend Port: 80
      vi. Backendpool: labbackendpool
      vii. HealthProbe: labhealthprobe
      viii. Leave other fields with their default values and click **OK**.

5. Create another virtual machine in the same virtual network:
   a. Go to the portal and create a virtual machine.
      i. VM Name: labvm
      ii. Region: Central US(same as vmss and Vnet)
      iii. Image: Windows server 2019 Datacenter
      iv. Size: Standard_B2ms
   b. Go to networking blade.
      i. Virtual Network: labvnet (should be same as VMSS)
      ii. NIC network Security Group: Advanced
      iii. Configure network security group: lbnsg (same as VMSS)
      iv. Load Balancing: No
      v. In Monitoring, set Boot Diagnostic: off
   c. Click **Create.**

6. Install IIS:
   You have to install webserver in each instance of VMSS, and you will be using labvm to connect both instances of VMSS.
   a. Connect to the labvm by RDP file with the credentials provided during creation.
   b. Go to the **Virtual Machine scale set**, go to **instances**, note down the private IP of both the instances.
   c. From labvm Virtual machine, RDP to vmss instance 1.
   d. Type **Windows+R**, in Run prompt, type **mstsc**.

e. In the Connect window, provide the Private IP address of first instance and then provide credentials. You are now logged in instance 1.
f. Open windows Powershell in instance 1 VM and type the following commands to install IIS webserver:

```
# install IIS server role
 Install-WindowsFeature -name Web-Server -IncludeManagementTools
# remove default htm file
 remove-item  C:\inetpub\wwwroot\iisstart.htm
 # Add a new htm file that displays server name
 Add-Content -Path "C:\inetpub\wwwroot\iisstart.htm" -Value $("Hello World
from " + $env:computername)
```

g. Repeat the same set of steps for the other instances of Virtual machine scale set.

7. Test the Load Balancer:
   a. From the load balancer, note down the Public IP of load balancer.
   b. Hit the Public IP in your browser, you will find the response coming from one of the instances of VMSS.
   c. Try the force Refresh by pressing **ctrl+F5**, you will find the response coming from the other instances of VMSS.

8. Delete the Resources:
   a. When no longer needed, delete the resource group, scale set, and all related resources.
   b. To do so, select the resource group for the scale set and then select **Delete**.

1.4 Troubleshooting

S. No	Problem	Solution
1	Not able to connect to VM	Check if RDP port 3389 is enabled, if not, add inbound rule to allow traffic to port 3389
2	Load Balancer URL not working	• Ensure to upgrade all instances of VMSS after connecting it to a load balancer • Check if webserver is installed on VMSS Instances
3	VM are in unhealthy state	Lower size VMs are generally not supported in Load Balancing, try changing the size of VM

1.5 Supporting References

Refer the below link for additional information:

1. https://docs.microsoft.com/en-us/azure/load-balancer/quickstart-load-balancer-standard-public-portal